Practice Project Overview

Exploring socioeconomic trends in US counties

In this project, you will embark on an exploration and analysis of demographic and economic indicators across various counties in the United States using the US census demographic data in Tableau public. As you leverage the strengths of data visualization, you will gain valuable insights into population distribution, income levels, employment patterns, and commuting trends.

Project scenario:

Imagine you're a researcher at a regional planning agency. You need to analyze the US census demographic data to understand the employment and unemployment rate, poverty rate, and income rate per capita for each county and state.

Through this project, you'll use Tableau public to:

- **Build hierarchies:** Organize data hierarchically based on geographic location.
- **Perform data modeling:** Create calculated fields to derive insightful metrics and potentially combine additional datasets.
- **Create basic visualizations:** Develop scatter plots, bar charts, and maps to visualize key findings.
- **Implement LOD and complex calculations:** Leverage LOD expressions and perform calculations for deeper analysis.
- **Use tooltips:** Enhance interactivity by providing additional information on hover.
- **Construct a final dashboard:** Compile individual visualizations into a cohesive dashboard for effective storytelling.

By completing this project, you'll gain hands-on experience with exploring real-world data and communicating insights through data visualization in Tableau public, while understanding the limitations of the free version compared to the full Tableau desktop.

About the data set:

This data set, derived from the American Community Survey (ACS) conducted by the U.S. Census Bureau in 2017, provides demographic and socio-economic information at the county level. It offers valuable insights into population characteristics, economic activity, and commuting patterns across various counties in the United States.

This public data set is available on the Kaggle website as *US census demographic data* under the *CC0: Public domain* license.

Hold the CTRL key while you click to download each of the following files:

acs2017_county_data.csv

acs2017_census_tract_data.csv

These datasets have 37 fields, 3220 rows and 37 fields, 74001 rows respectively, providing you with a comprehensive set of data to explore and analyze.

Key variables:

- **Geographic identifiers:**
  - CountyID: Unique identifier for each county.
  - State: State abbreviation for the county's location.
  - County: Name of the county.
- **Demographic data:**
  - TotalPop: Total population of the county.
  - Men, Women: Number of males and females residing in the county.
  - Hispanic, White, Black, Native, Asian, Pacific: Population distribution based on different ethnicities and races.
  - VotingAgeCitizen: Number of citizens eligible to vote in the county.
- **Economic data:**
  - Income: Median household income.
  - IncomeErr: Margin of error for the median household income.
  - IncomePerCap: Per capita income.
  - IncomePerCapErr: Margin of error for the per capita income.
  - Poverty: Percentage of the population living below the poverty line.
  - ChildPoverty: Percentage of children living below the poverty line.
- **Occupation and industry data:**
  - Professional, Service, Office, Construction, Production: Percentage of the employed population in various occupational categories.
- **Commuting data:**
  - Drive, Carpool, Transit, Walk, OtherTransp: Percentage of workers using different commuting methods.
  - WorkAtHome: Percentage of workers who work from home.
  - MeanCommute: Average commute time in minutes.
- **Employment data:**
  - Employed: Total number of employed individuals.
  - PrivateWork, PublicWork: Number of individuals employed in private and public sectors, respectively.
  - SelfEmployed: Number of self-employed individuals.
  - FamilyWork: Number of individuals working in their family business.
  - Unemployed: Number of unemployed individuals.

You will use the following two data sets to show data modeling.

- acs2017_census_tract_data.csv: Data for each census tract in the US, including DC and Puerto Rico.
- acs2017_county_data.csv: Data for each county or county equivalent in the US, including DC and Puerto Rico.

The two files you have downloaded have the same structure, with just a small difference in the name of the id column. You may notice that counties are political subdivisions, and the boundaries of some have been set for centuries. On the other hand, census tracts are defined by the Census Bureau and will have a much more consistent size. A typical census tract has around 5,000 residents.

Tasks in the project:

Task 1. Building hierarchies

1.1 Create a hierarchy named *Geographic Hierarchy* using **County** and **State**.

Task 2. Simple data modeling

2.1 Build one-to-many relationships with **acs2017_county_data.csv** and **acs2017_census_tract_data.csv**.

2.2 A) Calculate a new field named *Employment Rate* by dividing **Employed** by **Voting Age Citizen** and multiplying by 100.

2.2 B) Calculate a new field named *Poverty Rate* by dividing **Poverty** by **Total Pop** and multiplying by 100.

Task 3. Creating basic visualizations:

3.1 Create a scatter plot to visualize the relationship between **Employment Rate** and **Poverty Rate**.

3.2 Create a packed bubble map using **Geographic Hierarchy** and **Total Pop**.

3.3 Create a bar chart to compare **Unemployment** across **Geographic Hierarchy**.

Task 4. Create LOD and complex calculations (using calculated fields):

4.1 Create a calculated field named *Average Income per County* using Level of Detail (LOD) calculations.

4.2 Create a calculated field named *Poverty rate difference* by subtracting **Poverty** from **Child Poverty**.

Task 5. Using tooltips:

5.1 Enhance your map visualizations by adding tooltips that display additional information on hover, such as *Average Income per County*, *Poverty rate difference*, and *Income per capita*.

Task 6. Creating a final dashboard:

6.1 Create a dashboard arrange the created visualizations (scatterplot, map, and bar chart) and calculated fields into a visually appealing and informative dashboard.

Task 7. Implementing dashboard actions:

7.1 Create a filter action that allows users to select a specific **County** on the map and see the corresponding scatter plot and bar chart for that county.

7.2 Implement a highlight action that emphasizes listings of a specific **County** on the map when hovering on a scatter plot.

Author(s)

Shubhra Das